INCH-POUND

MIL-E-29581(AS)
21 JUNE 1991

MILITARY SPECIFICATION

EARCUP, UNIT ACTIVE NOISE REDUCTION SYSTEM GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for an Active Noise Reduction (ANR) unit installed in helmet earcups. The ANR unit is designed to operate with audio signals from the aircraft Intercommunication Set (ICS). The objective of the ANR unit is to reduce the levels of acoustic noise to which the crewmember is exposed, thereby providing an increase in the intelligibility of ICS audio signals, increased comfort, and reduced crewmember fatigue and hearing damage risk.

1.2 Part Numbers. Part numbers for ANR systems furnished under this specification shall be as follows: M29581/X-Y with "X" being the specification sheet number, and "Y" being the ANR system or parts of the system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, Systems Engineering Standardization Department (Code 53) Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5965

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SPECIFICATIONS

Military

MIL-S-5002	Surfaces Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-E-5400	Electronic Equipment, Aerospace General Specification For
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment
MIL-E-18239	Earphone H-87B/U
MIL-S-19500	Semiconductor Devices General Specification For
MIL-A-23595	Amplifier, Audio Frequency
MIL-A-23595/1	Amplifier, Audio Frequency, AM-3597C/A
MIL-M-26542	Microphone, Dynamic, General Specification for
MIL-M-26542/2	Microphone, M87/AIC, M26542/2-01 M26542/2-02, 26542/2-03
MIL-M-38510	Microcircuits, General Specification For
MIL-P-55110	Printed Wiring Boards General Specification For

STANDARDS

Military

MIL-STD-411	Aircrew Station Signals
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-750	Test Methods for Semiconductor Devices
MIL-STD-756	Reliability Modeling and Prediction
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810	Environmental Test Methods and Engineering guidelines
MIL-STD-976	Certification Requirements for JAN Microcircuits

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MIL-STD-1285	Marking of Electrical and Electronic Parts
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1772	Certification Requirements for Hybrid Microcircuit Facilities and Lines
MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	Calibration System Requirements

HANDBOOKS

Military

MIL-HDBK-217	Reliability Prediction of Electronic Equipment
MIL-HDBK-781	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA, 19111-5094.)

Naval Air Systems Command

AS-4613	Application and Derating Requirements for Electronic Components General Specification for

(Copies of AS-4613 may be obtained from the Naval Air Systems Command Headquarters, Specification and Records, Washington D.C. 20361-5110).

NA 13-1-6.7	Aircrew Personal Protective Equipment

(Copies may be obtained from the Naval Air Technical Services Facility, 700 Robbins Avenue, Philadelphia, PA 19111).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SD-6	Provisions Governing Qualification

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Non-Government publications. The following document forms a part of this specificaiton to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA 55-84 (ANSI S12.6-84) - Measurement of the Real-Ear Attenuation of Hearing Protectors, Method for (Application for copies should be addressed to the Acoustical Society of America, 335 East 45th Street, New York, NY 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification Sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. ANR systems furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids.

3.3 Design and Construction.

3.3.1 Materials, Processes, and Parts. Except as modified herein, parts, materials, and processes shall meet the requirements of MIL-P-11268, MIL-E-5400 and MIL-STD-454. The contractor shall submit for approval any parts, materials, and processes to be utilized in the construction of the ANR unit which do not comply with the above documents.

3.3.1.1 Electronics Assembly.

3.3.1.1.1 Quality Parts. The contractor shall use, to the maximum extent practical, established reliability (ER) capacitors, coils, filters, relays and resistors. Printed wiring assemblies shall be processed for compliance in accordance with MIL-P-55110 in order not to affect life and serviceability of assemblies.

3.3.1.2 Finishes.

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3.3.1.2.1 Metal parts. Surface treating of metal parts shall be in accordance with MIL-S-5002.

3.3.1.3 Connectors.

3.3.1.3.1 Earcup terminals. Earcup terminals shall be as specified in the detail specification sheet.

3.3.1.4 Solid State Components. The ANR unit shall employ solid state circuitry throughout. Microelectronics technology shall be used to the maximum extent practicable.

3.3.1.4.1 Microelectronics. Microelectronics, including film microcircuits, monolithic microcircuits, multichip microcircuits, hybird microcircuits, and discrete semiconductors shall be used whenever such use is feasible and compatible with other requirements of this specification. The following requirements shall apply to the microcircuits and semiconductors used in accomplishing the design of the ANR unit:

a. Semiconductors shall conform to MIL-S-19500 which includes the quality conformance requirements per MIL-STD-750.
b. Power transistors utilized shall not have junction temperatures in excess of 110°C for ambients up to 72°C.
c. Maximum internal temperatures of integrated circuits and small signal transistors shall not exceed 125°C under the ambient condition specified herein.
d. No germanium devices shall be utilized in this equipment.
e. The contractor, vendor, and/or supplying manufacturer shall obtain certification in accordance with the requirements of MIL-STD-976, and MIL-STD-1772 for the applicable device types, and MIL-M-38510. The criteria established for certification and qualification assures that microcircuits and semiconductors are manufactured under conditions which have been demonstrated to be capable of continuously producing highly reliable products.

3.3.1.5 Toxicity. Materials and coatings used in the design of the ANR unit and associated components shall be non-toxic in accordance with MIL-STD-454, Requirement 1.

3.3.1.6 Flammability. Materials which will support combustion shall not be used. Definitions and restrictions on the use of flammable materials shall be in accordance with MIL-STD-454, Requirement 3.

3.3.1.7 Encapsulation and embedment (potting). Selection of materials used in the earcup design shall meet the requirements of MIL-STD-454, Requirement 47.

3.3.2 Physical Characteristics.

3.3.2.1 Earcup Assembly. The ANR earcups shall be functionally and physically interchangeable with earcups presently used in the SPH-3C helmet. The earcups shall fit into the earcup retention system of the SPH-3C helmet, and shall fit properly into the helmet. Unless otherwise specified in the

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helmet procurement specification, dimensions of the earcups shall be as specified in the detail specification (see 3.1). Installation of the earcups shall be according to established maintenance procedures defined in NAVAIR Manual 13-1-6.7, and shall not require modification of the helmet assembly or earcup retention system. Earcups shall meet the requirements imposed upon current headgear. The earcup seal pad shall have a minimum service life of 6 months, and shall be replaceable by established maintenance procedures.

3.3.2.1.1 Rigid Earcups. The earcups shall be fabricated from A.B.S plastic or any equally suitable material. Unless otherwise specified the color shall be white. If the earcup is constructed of more than one piece, the pieces shall be permanently joined together. All seams shall be smooth.

3.3.2.2 Helmet cable assembly. The cable assembly used to connect the ANR system to the aircraft cable shall be in accordance with the detail specification sheet (see 3.1).

3.3.2.3 Bypass switch assembly. The bypass switch assembly shall be used to interrupt the ANR bias voltage to place the ANR unit in the active or bypass modes. The switch shall be mounted internal to the switch housing component of the helmet mounted boom microphone assembly, in accordance with the helmet specification or configuration drawing. The switch is considered part of the helmet cable assembly. Requirements for the switch are specified in the detail specification sheet for the helmet cable assembly.

3.3.2.4 Earphone. The earphone shall as a minimum meet the material characteristics of the H-87B/U earphones governed by MIL-E-18239, unless otherwise specified (see 3.1).

3.3.2.5 Electret boom microphone. Unless otherwise specified herein, the electret boom microphone element shall meet the requirements of the design and construction, and performance sections of microphone specification MIL-M-26542 and associated detail specification sheet MIL-M-26542/2. The preamplifier for the electret microphone shall meet the electrical characteristic requirements of preamplifier specification MIL-A-23595, and associated detail specification sheet MIL-A-23595/1. The electret microphone preamplifier shall be an integral part of the electret microphone. The output impedance of the electret microphone element and the input impedance of the microphone preamplifier are exempt from the requirements stated in MIL-M-26542 and MIL-A-23595.

3.3.2.6 Marking. The earcups containing the ANR unit shall be marked in accordance with MIL-STD-1285. Only the following information shall be included:

Contractor and Government Entities (CAGE)
Contract Number
Serial Number
Part number
Specify Right or Left earcup

3.3.2.7 Deleted.

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3.3.2.8 Soldering. Soldering shall be in accordance with MIL-STD-2000, Task G.

3.3.2.9 Workmanship. The assembled ANR units shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, functioning, and appearance. Cleaning of soldered connections shall be in accordance with 3.3.2.8. All other cleaning shall be in accordance with MIL-STD-454, requirement 9. There shall be no evidence of loose contacts, poor or improper molding or fabrication, damaged or improperly assembled contacts, peeling, flaking or chipping of plating or finish, mechanical damage due to testing environments, nicks or burrs of metal parts of surfaces, or improper or incorrect marking.

3.3.3 Interface Definition.

3.3.3.1 ICS Interface. The ANR unit shall be designed so that audio signals and DC electrical power shall be obtained by connecting the helmet cable directly to the aircraft cable. Adapter cables shall not be used. The helmet cable shall meet the requirements of 3.3.2.2.

3.3.3.2 Earcup and earphone interface. The earcup shall contain a male connector whose shell and pins do not protrude beyond the surface of the earcup. The earcup connector shall directly mate to the helmet cable connector, and shall hold the helmet cable connector secure via the helmet cable mounting screws. Removal and replacement of the earcup assembly shall be possible without the need for special tools.

3.3.3.3 Boom Microphone Interface. The electronics of the ANR unit shall connect electrically to the electret boom microphone via the helmet cable assembly. The electret microphone shall be physically interchangeable with the existing M-87/AIC microphone.

3.4. Performance Characteristics.

3.4.1 Acoustic Performance.

3.4.1.1 Earphone Sensitivity. When tested in accordance with 4.6.3.1, and with the ANR unit in the active or bypass modes, the normal sea level sound pressure level vs frequency characteristics of the earphone shall, as a minimum, be 73 dB Sound Pressure Level (SPL) at 250 Hz, and 85 dB SPL at 400 Hz, when an audio signal level of 1 milliwatt (1 mW) RMS, at each frequency is applied to the audio input terminals of the helmet cable assembly connector.

3.4.1.2 Frequency Response (Sea level). The frequency response shall be measured in accordance with 4.6.3.2 with the ANR system in the ON (active) and OFF (bypass) modes. This test shall be performed as part of the baseline tests, and shall be used for reference when designated in other paragraphs of this specification. A frequency response curve of the H-87B/U earphone currently used in aircrew helmets is provided in section six for informational purposes only.

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3.4.1.2.1 Frequency Response (Altitude). The normal sound pressure level vs frequency characteristics of each earphone shall not vary more than 4 dB from normal sea level performance when subjected to an ambient pressure of 13.75 inches of mercury (20,000 ft), nor more than 12 dB at an ambient pressure of 3.4 inches of mercury (50,000 ft).

3.4.1.3 Channel Balance. When tested in accordance with 4.6.3.3, the sound pressure difference between each earphone shall not be greater than 4 dB when measured at 1 KHz.

3.4.1.4 Linearity. When tested in accordance with 4.6.3.4, the transfer function dB SPL Out/dB PWR in (ICS Audio) shall be linear within ±2 dB from a SPL of 75 dB until the point at which the earphone produces a SPL of 130 dB.

3.4.1.5 Overload Protection. When in the active mode the ANR design shall limit the amount of cancelling noise to safe levels during extreme noise transients. Under no conditions, including instability of the ANR system, shall the earphone output sound pressure level exceed 130 dB SPL when tested in accordance with 4.6.3.5.

3.4.1.6 Total Attenuation. Total attenuation of a flight helmet fitted with an ANR system is the attenuation of the helmet when the ANR system is in the active mode (combined passive and active attenuation). The total attenuation shall not be less than the values specified in the detailed specification sheet during all operating scenarios anticipated during use, such as altitude variations, ambient noise levels, and electromagnetic interference. The ANR units shall provide the attenuation specified in the detail specification sheets when subjected to ambient noise levels simulating the overall sound pressure level of the aircraft.

3.4.1.6.1 Passive Attenuation. Passive attenuation of a flight helmet fitted with an ANR system is the attenuation of the helmet with the ANR system in the bypass mode. Minimum passive attenuation (i.e., ANR "OFF") determined in accordance with ANSI S12.6-1984, shall be as specified in the detail specification sheets.

3.4.1.7 Intelligibility. When the ANR sytem is in the active mode and tested in accordance with 4.6.3.7, the mean intelligibility scores obtained utilizing the Modified Rhyme Test (MRT) or the Tri-Word Modified Rhyme Test (TMRT) shall not be less than the percentage correct scores shown in Table I.

Table I. SPEECH INTELLIGIBILITY (Pink Noise Spectrum)

Overall Noise Sound Pressure Levels

Percent correct Intelligibility scores	75 dB	95 dB	105 dB	115 dB
	95	90	85	80

3.4.1.8 Undesired Sounds. The ANR unit shall not produce any undesired sounds when tested in accordance with 4.6.3.8.

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3.4.1.9 Noise Floor. When the ANR system is in the active mode and tested in accordance with 4.6.3.9, the level of the noise floor shall not exceed 35 dBA.

3.4.1.10 Deleted.

3.4.1.11 Alternate Earcup sound attenuation. At the discretion of the testing activity, this test shall be performed in accordance with 4.6.3.11 on each ANR unit prior to the performance of any test specified herein. The active attenuation value shall be recorded for future reference. This test shall also be conducted after the completion of any test specified herein to determine if the attenuation value has changed. If a change has occured the new attenuation value shall be recorded, and the test which caused the change shall be noted. The purpose of this test is to identify when a change in the attenuation value has occured, and the environment which caused the change. No pass/fail criteria have been established for this test, since the ANR units are subjected to the active and passive attenuation test of 3.4.1.6.

3.4.2 Electrical Performance. The ANR unit shall meet the following performance requirements when in the active or bypass modes, unless specifically stated.

3.4.2.1 Audio input signal voltage range. When tested in accordance with 4.6.4.1, the ANR unit shall meet the distortion requirements of 3.4.2.12.1 with an audio input signal voltage and frequency range within the limits specified in the detail specification sheet. The nominal audio input signal for testing purposes shall be as specified in the detail specification sheet (see 3.1).

3.4.2.2 DC Normal Bias voltage. The DC bias voltages shall conform to MIL-STD-704 DC Normal Operating Characteristics. Voltage levels shall be as specified in the detail specification sheet.

3.4.2.3 DC Low voltage conditions. Low voltage levels shall be as specified in the detail specification sheet (see 3.1). When tested in accordance with 4.6.4.3, the ANR unit shall remain in the active mode until the DC bias voltage level has decreased below the low voltage specified in the detail specification sheet, at which point it is permitted for the ANR unit to automatically switch to the bypass mode. It is permissible for the ANR unit to remain in the active mode when the DC bias voltage level is below the low voltage specified in the detail specification sheet. The ANR unit shall return to and remain in the active mode when the DC bias voltage level returns to the low voltage specified in the detail specification sheet.

3.4.2.4 DC Abnormal Operation. When in the active mode or the bypass mode and tested in accordance with 4.6.4.4, the ANR unit shall not be damaged due to transients or overvoltage occurring on the DC power bus as indicated in MIL-STD-704 Envelope of Voltage Transient for 28 volts nominal DC system and, Limits for DC Overvoltage or Undervoltage for 28 Volts Nominal DC System. The unit shall resume normal operation within a minimum of 2.0 seconds after power has returned to normal operating characteristics. The audio signal shall not be affected during performance of this test.

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3.4.2.5 Reverse Polarity Protection. When tested in accordance with 4.6.4.5, the ANR unit shall not be damaged by connection to the DC bias voltage regardless of polarity, with the ANR unit in the active or bypass mode.

3.4.2.6 Power Consumption. The ANR system (see 6.1) shall meet the requirements specified herein and in the detail specification sheet when operated with the DC bias and test circuit defined in the detail specification sheet.

Note: The ANR system power includes the power necessary for operation of both ANR units and the electret microphone.

3.4.2.7 Audio Modulation. Undesired modulation on the microphone line, caused by the ANR system shall not impair communications. When in the active mode and tested in accordance with 4.6.4.7, the ANR unit shall not impose an audio modulation of greater than 3 millivolts RMS between 100 Hz and 7000 Hz, on the DC bias power supply.

3.4.2.8 Electrical Short-Circuit Protection. When tested in accordance with 4.6.4.8, the ANR unit shall not be damaged by electrically shorting the boom microphone, and the audio input terminals.

3.4.2.9 Warm-up Time. Under any combination of conditions defined herein and tested in accordance with 4.6.4.9, the unit shall operate and meet all requirements of this specification within a minimum of 2.0 seconds after applying the nominal DC bias voltage specified in 3.4.2.2 to the ANR unit. The audio signal shall not be interrupted or degraded during this test.

3.4.2.10 Impedance. When in the active or bypass mode and tested in accordance with 4.6.4.10, the input impedance of the audio input terminals of each ANR unit shall be as indicated in the detail specification sheet (see 3.1). At any frequency between 100 Hz and 7000 Hz, the impedance shall not vary from the value at 1000 Hz by more than 20%.

3.4.2.11 Stability. When tested in accordance with 4.6.4.11, the ANR unit shall be stable and shall show no evidence of regeneration or oscillation.

3.4.2.12 Distortion.

3.4.2.12.1 ANR operation. When tested in accordance with 4.6.4.1, the Total Harmonic Distortion (THD) shall not exceed 2.5% between the fundamental frequencies of 300 Hz and 7000 Hz.

3.4.2.12.2 Bypass (ANR off) operation. When tested in accordance with 4.6.4.1, the THD shall not exceed 1.8% between the fundamental frequencies of 300 Hz and 7000 Hz.

3.4.2.13 Reliability Assurance Tests and Procedures. The manufacturer shall subject each unit in the production lot to the reliability assurance tests and procedures test as specified in 4.6.6.

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TABLE II High Levels RS03 Signals.

Frequency 1/ (MHz)	Modulation Type	Pulse Width	Pulses/sec	Power 2/ Density (mW/cm^2)	Field 3/ Strength (V/M)
0.014-850	AM 4/	90% 1 KHz sine wave		0.66 5/	50
215-225	PAM 6/	10 usec	200	10.6 7/	200
215-225	PAM	200 usec	200	10.6	200
400-450	PAM	4 usec	300	10.6	200
850-940	PAM	10 usec	1000	10.6	200
850-940	PAM	125 usec	300	10.6	200
940-1215	PAM	10 usec	1000	10.6	200
1215-1365	PAM	10 usec	3000	10.6	200
1365-2900	PAM	1 usec	1000	10.6	200
2900-3100	PAM	3 usec	1000	10.6	200
2900-3100	PAM	70 usec	200	10.6	200
3100-3500	PAM	10 usec	3000	10.6	200
3100-3500	PAM	50 usec	3000	10.6	200
3500-9000	PAM	1 usec	1000	10.6	200
9000-14000	PAM	0.2 usec	1000	10.6	200
14000-18000	PAM	0.2 usec	50000	10.6	200
33000-33400	PAM	0.2 usec	2000	0.106	20

Notes:
1. Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.
2. Peak power density for PAM signals is defined as the power density as measured with an average power measuring device if the generator is run CW: i.e., average power density within the pulse envelope.
3. Volts per Meter equivalent shown for reference only.
4. Amplitude modulated.
5. Average power density.
6. Pulse amplitude modulated.
7. Peak power density.

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3.4.2.14 Electromagnetic Interference (EMI). The equipment shall comply with the requirements of MIL-STD-461 for Class A1b equipment as modified by this specification. These requirements shall be met in both operating modes.

3.4.2.15 Susceptibility Definition. The following criteria shall be applied to determine whether the ANR unit is susceptible to EMI. Any repeatable, non-commanded change in the equipment's modes of operation shall be considered an indication of susceptibility. This shall apply whether the change is transient (momentary) or non-transient (permanent) in nature. The inability to change modes of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility. During and after completion of the tests there shall be no degradation in the performance requirements of 3.4.1.6, 3.4.2.12 and 3.4.2.16, except that the distortion shall not exceed 12% during the test.

3.4.2.15.1 Radiated and Conducted Susceptibility. The ANR unit shall be capable of passing the following emission and susceptibility test requirements of MIL-STD-461 when tested in accordance with 4.6.4.12.

a. CS06

b. RS01

c. RS02 (Power Frequency)

d. RS03 (The field levels and modulations shall be as specified in Table II, RS03 signals).

3.4.2.16 Gain. The difference between the SPL emitted by the earphone when switched from the bypass mode to the active mode shall be +2, -0 dB SPL when tested in accordance with 4.6.4.13.

3.4.2.17 Audible warning signals. When tested in accordance with 4.6.4.14, the ANR system shall not degrade transmisson of audible warning signals as defined in MIL-STD-411, from the ICS system to the earphones.

3.4.3 Environmental Conditions. When tested in accordance with 4.6.5, the equipment shall operate as required under any of the environmental service conditions, or any probable combination thereof, as specified in MIL-E-5400 for Class 1a equipment, except as modified herein.

3.4.3.1 Vibration. When tested in accordance with 4.6.5.1, the ANR unit shall operate in accordance with the requirements specified herein when subject to random and sinusoidal vibration requirements set forth in MIL-STD-810.

3.4.3.2 Temperature/Altitude. When tested in accordance with 4.6.5.2, the ANR unit shall comply with the temperature/altitude requirements set forth in MIL-E-5400 for Class 1a equipment.

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3.4.3.3 Humidity. When tested in accordance with 4.6.5.4, the ANR unit shall withstand the effect of humidities up to 100 percent, including conditions wherein condensation takes place in and on the equipment. The equipment shall withstand the above conditions during operating and non-operating conditions.

3.4.3.4 Mechanical Shock.

3.4.3.4.1 Functional Shock. The ANR unit shall not suffer damage or subsequently fail to provide specified performance when subject to 18 impact shocks as specified in 4.6.5.3.1.

3.4.3.4.2 Crash Hazard. The ANR unit shall withstand 12 impact shocks as specified in 4.6.5.3.2. Bending and distortion shall be permitted; however, no parts shall protrude beyond the envelope dimensions of the item under test. In addition, no parts shall break and propel from the ANR unit.

3.4.3.5 Acoustical Noise. When subjected to a pink noise environment of 140 dB SPL as specified in 4.6.5.5, the ANR unit shall show no signs of damage.

3.4.3.6 Fungus. When tested in accordance with 4.6.5.6, the ANR unit shall provide no nutrients in material, coating or contaminant which would support fungal growth. Only inherently fungus resistant grades of materials, per Requirement 4, MIL-STD-454, shall be used. Evidence provided to the testing activity which certifies the use of fungus inert materials may eliminate the necessity to perform the test indicated in 4.6.5.6.

3.4.3.7 Sand & Dust. When tested in accordance with 4.6.5.7, the ANR unit shall withstand, in both operating and non-operating conditions, exposure to wind blown dust (fine sand) particles.

3.4.3.8 Salt Fog. After completion of the salt fog test of 4.6.5.8, the earcups and ANR unit shall show no signs of corrosion.

3.5 Reliability and Maintainability.

3.5.1 Maintainability. Each helmet earcup containing an ANR unit shall be field installable and removable into the current rotary wing headgear by Organizational level maintenance facilities using common shop equipment. Failure of an ANR unit shall be determined at the Organizational level. The failed ANR unit shall be nonrepairable, except for replacement of the earcup seals, and the cable assembly.

3.5.2 Reliability program plan. The Reliability Program Plan is to have the signature approval of the contractor's reliability and program management. The reliability plan, as approved by the procuring activity, shall become the basis for contractual compliance. The plan shall provide for and define the following tasks and techniques to be used:

Management	Organization responsibilities, authority, relationship to program tasks, technical direction and control techniques.

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Schedule	Time phased plan to accomplish all reliability program tasks and to identify their relationship to engineering design and other program milestones (e.g., design, laboratory and flight tests, manufacturing release, design review, procurement release).
Analytical Tasks	The plan shall define the contractor's responsibilities and techniques used to fulfill analytical tasks such as reliability predictions, reliability stress analysis, etc.
Control Tasks	The plan shall define and describe procedures and techniques to be utilized in controlling the reliability design implementation and program. Specific attention shall be given to the parts control program and practices.
Evaluation Tasks	The plan shall define the scope and implementation disciplines for the reliability test program, procedures, plan, failure analysis, failure reporting, corrective action, etc.
Status Reports	The plan shall define in detail both formal and informal status reporting.

3.5.2.1 Records and Reports. The contractor shall maintain verifiable objective evidence that all reliability program tasks have been performed in accordance with specified requirements. Contractor internal records and reports such as engineering logbooks, charts, materials and processes selection criteria, failure data and reports, design review results, etc., shall be maintained in the contractor's format and made available for review upon request.

3.5.3 Mean Time To Replacement. The ANR unit shall have lower and upper test MTTR values of 25,000 and 50,000 hours respectively, when tested in accordance with MIL-STD-781 when operated in any mode and under worse case environmental conditions specified in paragraph 4.6.5.

3.5.4 Reliability stress analysis. The environmental condition of an airborne, rotary winged aircraft at +71°C plus internal rise shall apply for reliability predictions. Under this condition, the reliability prediction shall exceed the required theta 0 by a minimum ratio of 1.25 to 1. MIL-STD-756 and MIL-HDBK-217 shall be used as guidance for analysis with additional test data as it becomes available. The contractor shall conduct and include as part of the reliability stress analysis a part level stress analysis. The analysis shall be performed to verify derating is in accordance with AS-4613, Class A derating requirements. Part level stresses shall then be used for completion of a parts stress reliability prediction in accordance with MIL-HDBK-217 section 5.1. Additionally, the surface temperature of the potted assembly shall not exceed 40°C during normal operating conditions. Junction temperatures of semiconductor devices shall not exceed 110°C.

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4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Qualification inspection (see 4.4)

Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph(s) of this specification.

4.3.1 Test equipment. All instruments used in performing inspections required by this specification shall be maintained and calibrated in accordance with MIL-STD-45662. The intervals between calibration shall be no greater than one year.

4.4 Qualification inspection. Qualification inspection shall be performed in accordance with the provisions governing qualification (SD-6). The qualification inspection shall consist of all Group I tests specified in TABLE IV (see latter portion of this document). The qualification inspection sample shall consist of six ANR systems for which qualification is desired. Group IA inspections shall be performed on each of the sample items. Following Group IA tests, at least 2 samples shall be submitted to Group IC tests. Upon completion of Group IC tests, Group IB tests shall be performed on at least one unit which has been subjected to Group IA tests only, and one unit which has been subjected to Group IC.

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4.4.1 Packaging and marking Item for qualification must be separately packaged and forwarded to the activity responsible for qualification as designated in the letter of authorization. The sample shall in all respects be representative of the manufacturer's production item. Samples shall be plainly marked in accordance with 3.3.2.6 except the items shall also be marked (Qualification Samples) and shall include the applicable specification number, and shall reference the letter authorizing the tests.

4.4.2 Final acceptance Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Acceptance tests shall consist of Group II and Group III tests in TABLE IV.

4.4.3 Retention of qualification. Contractors wishing to remain on the Qualified Products List (QPL) shall submit to the preparing activity at intervals no greater than three years, DD Form 1718, Certification of Qualified Products. At the discretion of the qualifying activity, six (6) Qualification samples shall be submitted to the QPL agent for verification of design requirements. As a minimum, the QPL agent shall perform the Group I tests noted in TABLE IV, to allow retention of qualification, and any other tests deemed necessary to insure that the basic design requirements for the ANR systems are still being met. The contractor shall submit production units (or units representing production) as Qualification samples.

4.5 Quality conformance inspection. Quality conformance inspection for this specification shall consist of the tests of Group II and Group III of TABLE IV unless otherwise specified. Group II quality conformance tests shall be performed on each individual unit prior to submission to the Government for acceptance. Units failing the quality conformance test may be discarded or reworked to correct defects and resubmitted for acceptance.

4.5.1 Production lot sampling. All ANR systems submitted for inspection as a production lot shall be manufactured using identical components, raw materials and processes. All materials shall be from the same incoming lots. All Group III tests specified in TABLE IV shall be performed on sample units pulled from the production lot which has already passed the Group II inspection. The sample units shall be chosen at random under the supervision of the cognizant Government inspector. Sample sizes for Group III sample testing shall be as specified in TABLE III.

TABLE III. Sample plan for Group III sample testing.

Number of ANR units in contract	Total number of ANR units to be tested
1 thru 250	3
251 thru 500	6
501 thru 750	9
751 thru 1000	12
1001 thru 1250	15
1251 thru 1500	18
1501 or more	21

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4.5.1.1 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defect, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.5.1.2 Disposition of sample units. Sample units which have passed the Group III inspection may be delivered on the contract or purchase order if the lot is accepted and the deliverable sample units are still within specified electrical and acoustical tolerances, and meet the workmanship requirements of 3.3.2.9 In addition, the cable assembly shall not show signs of physical damage.

4.5.1.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found, by lot, contract number, and serial number, and the method of correcting them.

4.5.2 Laboratory facilities. Contractors not having laboratory testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the qualifying activity.

4.6 Methods of examination and test.

4.6.1 Test conditions. Unless otherwise specified, all inspections shall be performed in accordance with the following test conditions:

4.6.1.1 Acoustical measurement test arrangement. The ANR unit shall be removed from the helmet assembly unless otherwise specified. The complete ANR earcup (foam, padding), shall be placed over the ear of an artificial head. A "Kemar" manikin with a "Zwislocki" coupler or equivalent will be used. The artificial head shall be fastened in a horizontal position. A force of 1 kilogram shall be applied to the earcup. The test circuit arrangement is shown in Figure 1. The measurements are to be made in a reverberant or semi-reverberant room with ambient noise in a pink noise spectrum, 40 Hz to 10 KHz, at a level of 125 dB SPL. The measuring microphone shall be connected to a microphone preamplifier, and a measuring amplifier, from which direct measuring of sound pressure levels can be displayed.

4.6.1.2 Environmental conditions.

a.	Temperature	Room ambient (25 ± 10°C)
b.	Altitudes	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient, up to 90 percent relative humidity

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4.6.1.3 Electrical measurement test arrangement. All tests shall be performed using the DC bias circuit shown in the detail specification sheet. Electrical connection of the ANR unit to the test circuit shall be made via the helmet cable.

4.6.2 Visual and mechanical examinations. The ANR unit shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements.

4.6.2.1 Soldering. Soldering shall be examined in accordance with 3.3.2.8.

4.6.3 Acoustical performance tests. The acoustical test conditions shall be as specified in 4.6.1. Unless specifically stated, the nominal audio input signal and DC bias voltage range shall be as specified in 3.4.2.1 and 3.4.2.2 respectively. All audio signals and electrical power connections to the ANR unit shall be made via the helmet cable connector. All measurements of audio signals and electrical power shall be made at the input terminals of the helmet cable connector when possible. The ANR unit shall meet the following requirements when in both the active and bypass modes, unless specifically stated.

4.6.3.1. Earphone Sensitivity. The earcup shall be configured as specified in 4.6.1.1, except there shall be no ambient noise. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The ANR unit shall be placed in the bypass mode. An AC current meter shall be connected in series with the audio signal generator. The signal generator shall be set to a frequency of 250 Hz. The output of the audio signal generator shall be increased until the product of the audio input current and voltage is equal to one milliwatt (1 mW). The output sound pressure level of the earphone shall be recorded. The ANR system shall be switched to the active mode, and the audio signal level adjusted to a level of 1 mW. The output sound pressure level of the earphone shall be recorded. The frequency of the signal generator shall be changed to 400 Hz, and the test repeated.

4.6.3.2 Frequency response. The earcup shall be configured as specified in 4.6.1.1, except there shall be no ambient noise. The ANR unit shall be placed in the bypass mode. An AC current meter shall be connected in series with the audio signal generator. The output of the audio signal generator shall be increased until the product of the audio input current and voltage is equal to one milliwatt (1 mW). The frequency of the signal generator shall be varied from 100 Hz to 7000 Hz, while recording the output sound pressure level of the earphone. The test shall be repeated with the ANR unit in the active mode, with the nominal DC bias voltage specified in 3.4.2.2. For frequency response at altitude, subject the earphone to the ambient pressures specified in 3.4.1.2.1 during the measurement periods.

4.6.3.3 Channel Balance. The data obtained during the frequency response test of 4.6.3.2 for each earcup in the active mode, shall be analyzed to determine if the requirements of 3.4.1.3 have been met.

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4.6.3.4 Linearity. The earcup shall be configured as specified in 4.6.1.1, except there shall be no ambient noise. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The ANR unit shall be placed in the bypass mode. The frequency of the signal generator shall be set to 1000 Hz. While the audio input signal is varied from 0 to .82 Watts, the output sound pressure level of the earphone shall be monitored to determine compliance with 3.4.1.4. The test shall be repeated with the ANR unit in the active mode.

4.6.3.5 Overload protection. The earcup shall be configured as specified in 4.6.1.1, except there shall be no ambient noise. This test shall be performed in the active mode only, with the nominal DC bias voltage specified in 3.4.2.2. The frequency of the signal generator shall be set to 1000 Hz, or the broadband noise from 20 Hz to 10 KHz may be used. While monitoring the output SPL from the earphone, the output of the signal generator shall be increased to the limits specified in 3.4.2.1 to attempt to produce a SPL greater than 130 dB to determine compliance with 3.4.1.5.

4.6.3.6 Active and passive attenuation. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. Physical ear attenuation measurements of the combined active and passive attenuation shall be obtained on human test subjects wearing the helmet-integrated ANR system while situated in a semi-reverberant room into which is fed a broadband noise of at least 85 dB SPL. The subjects shall be fitted with a small microphone positioned at the entrance to the ear canal. A one-third octave band analysis of the microphone's output shall be made without the helmet-integrated ANR system being worn and with the system worn and the ANR unit switched "ON". The difference in sound pressure levels for the two conditions shall be considered the combined active and passive (total) attenuation of the helmet-integrated ANR system. The mean attenuation shall be computed for each one-third octave test band by averaging the attenuation values for a minimum of ten subjects. The test shall be repeated at higher ambient noise levels, as specified in the detailed specification sheet, to ensure performance of the ANR system is not degraded. Attenuation measurements at levels greater than 115 dB SPL shall be obtained on an artificial head.

4.6.3.7 Intelligibility. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The intelligibility measurements shall be made with the talker in quiet ambient (or utilizing recorded speech materials) and with the listeners in a broadband pink noise environment having the overall sound pressure levels specified in Table I. The speech level shall remain constant at 85 dB SPL for all four test conditions, as measured under the earcup (at the entrance to the subject's ear canal). Intelligibility scores shall be obtained on a minimum of six listeners wearing the helmet-integrated ANR system in both the active (ANR "ON") and bypass (ANR "OFF") modes. The talker shall use a comfortable, communicative voice level and equal vocal effort on the carrier phrase employed for the intelligibility test words.

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4.6.3.8 Undesired sounds. The ANR shall be placed in the active mode only, with a DC bias as specified in 3.4.2.2. The ANR unit shall be placed against a human ear, or arranged as specified in 4.6.1.1. The following parameters shall be varied to determine compliance with 3.4.1.8. The sequence and test combinations shall be at the discretion of the testing activity.

a. The ANR unit shall be turned OFF, and back ON.

b. The power supply shall be varied between the limits of 11.0 to 29.0 volts, and as specified in 3.4.2.4, and 3.4.2.5.

c. The unit shall be placed in the active mode, with a DC bias voltage as specified in 3.4.2.2. The input signal level shall be varied to the extremes specified in 3.4.2.1 at a frequency of 1 KHz.

d. During normal operation the test subject shall compress the earcups to break the seal.

4.6.3.9 Noise floor. The ANR earcups shall be configured as specified in 4.6.1.1, except there shall be no ambient noise. A short circuit shall be placed across the ICS audio input terminals of the cable assembly. The earphone output sound pressure level present in the earcups shall be recorded prior to placing the ANR unit in the active mode. The ANR unit shall be placed in the active mode with a DC nominal bias voltage as specified in 3.4.2.2, and the earphone output sound pressure level shall be recorded. The noise present during the active mode shall not be greater than 35 dBA when compared with the noise level recorded when in the bypass mode.

4.6.3.10 Deleted.

4.6.3.11 Alternate Earcup sound attenuation. The ANR unit shall be centered over an artificial voice which has the capability to measure the SPL it produces. The nominal DC bias voltage specified in 3.4.2.2 shall be applied to the unit under test, and the unit shall be placed in the bypass mode. The measuring microphone of the artifical voice shall be placed at the mouth opening of the artificial voice. The output of the measuring microphone shall be connected to a measuring amplifier from which direct reading of sound pressure levels can be displayed. The artificial voice shall be connected to a pink noise generator with a suggested bandwidth of 100 Hz to 600 Hz, and adjusted for an output of 80±5 dB SPL. This value shall be recorded as the bypass sound pressure level. The ANR unit shall be placed in the active mode, and the sound pressure level recorded. The difference between the SPL measured when in the bypass and active modes shall be recorded as the active attenuation provided. This value shall be compared with the previous value recorded to determine if a change in active attenuation has occurred.

4.6.4 Electrical performance measurements. All tests shall be performed using the DC bias test circuit shown in the detail specification sheet. The electrical test conditions shall be as specified in 4.6.1. Audio signals and electrical power connections to the ANR unit shall be made via the helmet

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cable connector unless specifically stated. Measurements of audio signals and electrical power shall be made at the input terminals of the helmet cable connector when possible. The ANR unit shall meet the following requirements when in both the active and bypass modes, unless specifically stated.

4.6.4.1 Audio input signal voltage range and Distortion. The Total Harmonic Distortion (THD) shall be measured via the acoustic output of the earphone and calculated using the procedure specified below. The ANR earcup shall be configured as specified in 4.6.1.1 except there shall be no ambient noise. The input of the spectrum analyzer, or distortion analyzer, shall be connected to the AC output of the measuring amplifier. The measuring amplifier shall be in its most sensitive range while recording data. The ANR units shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The audio input signal voltage level shall be set between the limits specified in 3.4.2.1, and the frequency varied to the limits specified in 3.4.2.1. The level of the audio input signal shall be changed, and the test repeated. The value of the audio input signal level chosen and the number of levels checked shall be at the discretion of the testing activity. As a minimum, the nominal and maximum input signal level and frequency range should be checked. The THD of the ANR unit under load conditions shall be determined using a distortion analyzer, or by using a spectrum analyzer to determine the amplitudes of the harmonics and using the following equation:

THD (Total harmonic distortion %) = [The square root of the summation of $(A_1/A_f)^2$] * 100

Where A_1 is the amplitude of the 1_{th} harmonic and A_f is the amplitude of the fundamental. All harmonics and subharmonics within the range of 300 to 10000 Hz shall be considered. The distortion shall be measured at fundamental frequencies of 300, 1000, 3000, 5000, 7000 Hz or as otherwise specified by the qualifying activity to insure compliance with the requirements of 3.4.2.12.

4.6.4.2 DC normal operation Voltage. During performance of all tests specified herein, the ANR units shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The DC bias voltage range shall be used during the performance of any tests when deemed necessary by the testing activity.

4.6.4.3 DC Low voltage conditions. The ANR unit shall be configured as specified in 4.6.1.1 except the use of a human test subject is also permitted in order to hear when a change in the operating mode has occurred. The presence of an ambient background noise is not necessary and may be omitted. The ANR unit shall be placed in the active mode. The ANR unit shall be powered with the maximum DC bias voltage level specified in 3.4.2.2. No audio signal shall be applied during this test. While monitoring the earphone of the ANR unit, the DC bias voltage shall be decreased until the ANR unit automatically switches to the bypass mode. The DC bias voltage shall be recorded to determine compliance with 3.4.2.3. The DC bias voltage shall be increased until the ANR unit returns to the active mode. The DC bias voltage shall be recorded to determine compliance with 3.4.2.3. The following test shall also be performed as a minimum with the DC bias voltage decreased to the low voltage condition specified in 3.4.2.3 while in the active mode only, to determine compliance with the electrical and acoustical performance requirements of this specification: 4.6.4.13.

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4.6.4.4 DC Abnormal operation. The ANR unit shall be configured as specified in 4.6.1.1 except the presence of an ambient background noise is not necessary and may be omitted. The output of a signal generator shall be set to the nominal audio input signal as specified in 3.4.2.1. The ANR unit shall be placed in the active mode. The ANR unit shall be powered with the DC bias voltage specified in MIL-STD-704 Envelope of Voltage Transient for 28 volts nominal DC system and, Limits for DC Overvoltage or Undervoltage for 28 Volts Nominal DC System. When the voltage has returned to the nominal DC bias voltage specified in 3.4.2.2, the output signal of the ANR unit shall be monitored in accordance with 4.6.4.9 to determine compliance with 3.4.2.4. After completion of this test, the following tests shall be performed as a minimum to determine compliance with the electrical and acoustical performance requirements of this specification: 4.6.3.11 and 3.4.1.2.

4.6.4.5 Reverse polarity protection. The output of a signal generator shall be set to the nominal audio input signal as specified in 3.4.2.1. The ANR unit shall be placed in the bypass mode. The maximum DC bias voltage for the ANR unit specified in 3.4.2.2 shall be applied in reverse polarity for 1 minute. The test shall be repeated with the ANR unit in the active mode. After completion of the test, the following tests shall be performed as a minimum to determine compliance with the electrical and acoustical performance requirements of this specification: 4.6.3.11 and 3.4.1.2.

4.6.4.6 Power consumption. The use of the test circuit provided in the detail specification sheets will ensure correct power levels are provided to the ANR unit under test.

4.6.4.7 Audio modulation. The ANR unit shall be placed in a pink noise environment as specified in 4.6.1.1. The boom microphone shall be removed, or shielded from the ambient noise as much as practical to reduce the audio signal imposed on the DC bias line caused by the microphone. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The output of a signal generator shall be set to the nominal audio input signal as specified in 3.4.2.1. The ANR unit shall be placed in the active mode. The DC input power lines of the ANR unit shall be monitored with a spectrum analyzer as defined in the detail specification sheet. The test may be substituted via use of a two station ICS set-up wherein the human or manikin, test subject (A), placed in a noise environment, transmits his audio signal to a human test subject (B), positioned in a quiet environment. Initially test subject (B) adjusts his audio level to a comfortable level. The boom microphone of test subject (A) and (B) should be removed or shielded from the ambient noise as much as practical to eliminate any audio signals being heard due to the microphones. Audio signals heard by test subject (B) is the result of current modulation caused by operation of the ANR system, and any interference induced by the ICS set-up. Test subject (B) subjectively determines if any undesirable audio signals are received.

4.6.4.8 Electrical short circuit protection. The ANR unit shall be powered with the nominal DC bias voltage specified 3.4.2.2, and shall be placed in the active mode. The audio input terminals shall be shorted for a period of 2 minutes. At the completion of 2 minutes, the short circuit shall

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be removed and the boom microphone terminals shall be shorted for 2 minutes. The following tests shall be performed as a minimum to determine compliance with the electrical and acoustical performance requirements of this specification: 4.6.3.11 and 3.4.1.2.

4.6.4.9 Warm-up time. The ANR unit shall be placed in the bypass mode, and configured as specified in 4.6.1.1. The measuring amplifier shall be in its most sensitive range while recording data. The interval timer, or digital storage oscilloscope, shall be configured to trigger upon the application of the nominal DC bias voltage to the ANR circuit, and stop upon detection of the noise cancelling signal. The ANR unit shall be switched to the active mode, and the test repeated. Detection of the acoustic noise cancelling signal emitting from the ANR earphone shall occur within 2 seconds.

4.6.4.10 Impedance. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The impedance shall be measured as defined in the detail specification sheet. The frequency of the input signal shall be varied between the limits of 100 Hz to 7000 Hz, to determine compliance with 3.4.2.10. The ANR unit shall be placed in the active mode, and the test repeated.

4.6.4.11 Stability. The ANR unit shall be powered with the nominal DC bias voltage specified in 3.4.2.2, and the units shall be placed in the active mode. The ANR unit shall be configured as specified in 4.6.1.1 except the presence of an ambient background noise is not necessary and may be omitted, and the use of a human test subject may be used if desired. The audio input terminals of the ANR unit and boom microphone shall be shorted independently for a minimum of 2 minutes, while the DC bias voltage is varied from 0.0 to 29.0 volts. The output signal shall be monitored and shall show no signs of oscillation during or after the test. The test shall be repeated with the audio input terminals and boom microphone input terminals terminated in an open circuit.

4.6.4.12 Radiated and Conducted susceptibility. The ANR units shall be tested in accordance with the applicable test procedures of MIL-STD-462, and shall meet the requirements of 3.4.2.15. The ANR units shall be powered with the nominal DC bias voltage specified in 3.4.2.2. The output of a signal generator shall be set to the nominal audio signal specified in 3.4.2.1. The output distortion shall meet the requirements of 3.4.2.12 when monitored as specified in 4.6.4.1. The gain shall be monitored in accordance with 4.6.4.13 and shall meet the requirements of 3.4.2.16. The DC bias voltage shall be decreased to initiate the automatic change to the bypass mode. The DC bias shall then be increased to initiate return to the active mode. The output of the ANR unit should be monitored during each initiation stage to determine compliance with 3.4.2.3. The attenuation shall be measured as defined in paragraphs 4.6.3.6 or 4.6.3.11 at the discretion of the testing activity. Noncompliance with the above requirements is an indication of susceptibility.

The distortion test defined above may be replaced with the following test at the discretion of the testing activity. While subjecting the ANR system (bypass and active modes) to the EMI environment, an intelligibility test consisting of a Tri-Word Modified Rhyme (TWMRT) test shall be performed as follows: The ANR system shall be configured on a Kemar manikin as defined in 4.6.1.1. A tape recorder shall be connected to the audio input terminals of

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the ANR system, and shall be chosen so as not to affect the results of the intelligibility scores. A tape containing the data for the TWMRT shall be placed in the tape recorder, and played during the intelligibility test. The speech level (dB SPL) shall be as specified in the intelligibility test 4.6.3.7 and measured as defined therein. The audio output of the microphone located internal to the Kemar shall be connected to the audio input terminals of a cassette tape recorder, via the microphone power supply, to record the TWMRT data being emitted from the earphones of the ANR system. Levels used during recording shall be monitored to ensure minimum distortion is added to the system. The intelligibility test shall be performed as defined in 3.4.1.7, however the ambient noise shall be minimized as much as possible, and the tape shall be played back to test subjects using helmets, or an ANR system placed in the bypass mode, chosen so as not to affect the intelligibility test results.

4.6.4.13 Audio signal gain. The frequency response test 3.4.1.2, shall be performed to determine conformance with the requirement of 3.4.2.16.

4.6.4.14 Audible warning signals. The audio signal gain test of 4.6.4.13 shall be performed with the exception that the audio input signals shall be those defined below. A programmable audio signal generator, capable of modulation should be connected to the audio input terminals of the ANR helmet cable assembly vice the signal generator defined for use in 4.6.4.13.

a. Audio signal sweep from 700 Hz to 1,700 Hz in 0.85 second. An interruption interval of 0.12 second may be used.
b. Fundamental frequency of 250 ± 50 Hz, interrupted at 5.0 ± 1.0 Hz with a 50 ± 10 percent on-off duty cycle.
c. Fundamental frequency of 400 Hz interrrupted at a rate of 1 to 10 Hz.
d. Fundamental frequency of 1600 Hz interrupted at a rate of 1 to 10 Hz.
e. Fundamental frequency of 900 Hz.
f. Fundamental frequency of 900 Hz plus 400 Hz signal interrupted at a rate of 0 to 1 Hz.
g. Fundamental frequency of 900 Hz plus 1600 Hz signal interrupted at a rate of 0 to 1 Hz.

4.6.5 Environmental tests. The environmental tests shall be conducted in accordance with the following procedures to demonstrate that the equipment meets the requirements of this specification and MIL-E-5400. When an operational check is desired, the unit shall be energized with the nominal DC bias voltage specified in 3.4.2.2 unless otherwise specified. Performance checks which require the unit to be energized at the lowest and highest operating voltage shall be energized with the voltage range specified in 3.4.2.2. The equipment shall provide satisfactory operation as specified herein before and within two hours after performing the humidity test and after the salt fog test. The ANR unit shall provide satisfactory operation after completion of each of the rest of the environmental tests, with the exception of the crash hazard test. A performance test is not required after completion of the crash hazard test. The following performance verification

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tests shall be performed as a minimum, to determine compliance with this specification: 3.3.2.9, 4.6.3.11, and 3.4.1.2. The active and passive attenuation test of 4.6.3.6 shall be performed as soon as feasibly possible after completion of all the environmental tests.

The sequence of the tests shall be as follows:

a. Mechanical shock
b. Vibration
c. Acoustical Noise
d. Temperature-altitude
e. Humidity
f. Fungus
g. Salt fog
h. Sand and dust

Units which are submitted to the humidity test shall not be subjected to the fungus or salt fog test. Units subjected to the fungus test shall not be submitted to the salt fog test. It is desirous to subject most items to tests (a), (b), and (c). Test experience has shown that climate-sensitive defects often show up more clearly after the application of shock and vibration forces. For test (a) only the functional shock portion of this test applies to multiple units, since the crash hazard portion may render the units inoperable.

4.6.5.1 Vibration. The equipment shall be subjected to the vibration test of MIL-STD-810, Method 514.3, Procedure I for Category 6 equipment and Figure 514.3-28, where:

F1 = 20 Hertz	L1 = 2.0 g
F2 = 40 Hertz	L2 = 2.5 g
F3 = 60 Hertz	L3 = 1.5 g
F4 = 80 Hertz	L4 = 1.5 g

Vibration shall be applied sequentially along each of three mutually perpendicular axes for one hour.

4.6.5.2 Temperature/Altitude. The equipment shall be tested in accordance with MIL-STD-810, Method 520.0, Procedure III, except that humidity and vibration environments shall be deleted. The performance checks specified in the various steps of this test shall consist of the frequency response test of 3.4.1.2. Delete steps 2 through 7 and substitute the following:

1. Step 2. With the equipment non-operating, adjust the chamber test conditions to -54°C and ambient pressure. The equipment temperature shall be stabilized and maintained for at least two hours.

2. Step 3. With the equipment non-operating, adjust the chamber temperature to -40°C and ambient pressure. These conditions shall be maintained throughout this step. After the equipment temperature has stabilized, the equipment shall be operated at the lowest specified DC input voltage. The equipment shall operate satisfactorily immediately following the warmup time specified in 3.4.2.9. The equipment shall be de-energized and restabilized at -40°C The above operational sequence shall be repeated two additional times.

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3. Step 4. With the equipment non-operational, adjust the chamber temperature to -40°C and the equipment temperature allowed to stabilize. The equipment shall be energized at the highest specified DC input voltage and the chamber pressure adjusted to simulate a maximum altitude as specified in the detail specification sheet. Upon reaching the specified altitude and while maintaining the specified chamber temperature, the equipment shall be checked for satisfactory operation.

4. Step 5. With the equipment non-energized, adjust the chamber temperature to -10°C and ambient pressure. After the equipment temperature has stabilized, the chamber door shall be opened and frost allowed to form on the equipment. The chamber door shall remain open long enough for the frost to melt, but not long enough for the moisture to evaporate. The chamber door shall be closed and the equipment operated at the highest specified input voltage to ascertain satisfactory operation immediately following the warmup time specified in 3.4.2.9. The equipment shall be energized and de-energized at least three times.

NOTE: The intent is to have frost form when the chamber door is opened. Should the relative humidity of the air be such that frost will not form, an artificial means shall be used to provide the relative humidity necessary to have frost form.

5. Step 6. With the equipment non-energized, adjust the chamber test conditions to ambient temperature and pressure. After the equipment temperature has stabilized, the ANR unit shall be energized with the nominal DC bias voltage specified in 3.4.2.2, and an operational and performance check shall be made.

6. Step 7. With the equipment non-energized, adjust the chamber test conditions to 70°C and ambient pressure. The chamber temperature shall be stabilized and maintained for at least 16 hours. At the conclusion of this time period, where practicable, the equipment shall be visually inspected for any deterioriation.

7. Step 8. With the equipment non-operating, adjust the chamber test conditions to 65°C and ambient pressure. After the equipment temperature has stabilized and while maintaining the chamber temperature, operate the equipment continuously for 4 hours at the highest specified input voltage. At the end of the test period the equipment shall be checked for satisfactory operation.

8. Step 9. With the equipment non-operating, adjust the chamber test conditions to 36°C. The equipment temperature shall be stabilized. The equipment shall be energized at the highest specified input voltage and the chamber pressure adjusted to simulate a maximum altitude as specified in the detail specification sheet. Maintain these test conditions for a period of 4 hours. At the end of the time period, and while maintaining the specified test conditions, the equipment shall be checked for satisfactory operation.

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9. Step 10. With the equipment non-energized, adjust the chamber temperature to 10°C. The equipment temperature shall be stabilized. The equipment shall be energized at the highest input voltage specified and the chamber pressure adjusted to simulate a maximum altitude as specified in the detail specification sheet. Maintain these test conditions for a period of 4 hours. At the end of the time period, and while maintaining the specified test conditions, the equipment shall be checked for satisfactory operation.

4.6.5.3 Mechanical shock.

4.6.5.3.1 Functional shock. The cushion of the earcup shall be removed, and the face of the earcup shall be mounted against the vibrating structure. The equipment shall be subjected to the functional shock test of MIL-STD-810, Method 516.3, Procedure I using the test parameters for flight equipment specified in Figure 516.3-1. The test shall consist of three shocks in opposite direction along each of three mutually perpendicular axes.

4.6.5.3.2 Crash hazard. The cushion of the earcup shall be removed, and the face of the earcup shall be mounted against the vibrating structure. The equipment shall be subjected to the crash hazard shock test of MIL-STD-810, Method 516.3, Procedure V using the test parameters for flight equipment specified in Figure 516.3-1. The test shall consist of two shocks in opposite directions along each of three mutually perpendicular axes. The unit shall be nonoperating during this test.

4.6.5.4 Humidity. The ANR unit shall not be mounted in the helmet during performance of this test. The equipment shall be subjected to the humidity test of MIL-STD-810, Method 507.2, Procedure III and Figure 507.2-3. The test duration shall be 10 cycles.

4.6.5.5 Acoustic noise. The ANR unit shall be mounted in the helmet assembly simulating the normal operating configuration during performance of this test. The equipment shall be subjected to the acoustic noise test of MIL-STD-810, Method 515.3, Procedure II. During functional checks, the ANR unit shall be energized with the nominal DC bias specified in 3.4.2.2. The following details apply:

1. Step 4 - The test level shall be 140 dB SPL and duration shall be 30 minutes.

2. Step 5 - Delete.

4.6.5.6 Fungus. The equipment shall be subjected to the fungus test of MIL-STD-810, Method 508.3. The test duration shall be 28 days. This test may be accomplished by test or analysis. If Group II materials as specified in Requirement 4 of MIL-STD-454 are used, the test shall be performed.

4.6.5.7 Sand and dust. The equipment shall be subjected to the sand and dust test of MIL-STD-810, Method 510.2, Procedure I. The following details apply:

1. Step 2 - The air velocity shall be 1750 ± 250 feet per minute.

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2. Step 5 - Chamber temperature shall be 63°C and the air velocity shall be 300 ± 200 feet per minute.

3. Step 8 - The equipment shall not be operated.

4.6.5.8 Salt fog. The equipment shall be subjected to the salt fog test of MIL-STD-810, Method 509.2, Procedure I. The test duration shall be 48 hours.

4.6.6 Reliability Assurance Tests and Processes. Prior to subjecting the ANR units to the tests specified herein, each unit in the production lot shall be subjected to the Environmental Stress Screening (ESS) as follows:

4.6.6.1 Environmental Stress Screening (ESS). Each ANR unit shall undergo a period of Environmental Stress Screening prior to final acceptance test. The ESS shall consist of vibration and thermal cycling. Any screen tailoring shall be in coordination with and at the descretion of the procuring activity.

Vibration Screening: All end items shall receive random vibration for 10 minutes in three axes (see the 6 grms, 20 to 2000 hertz profile of Figure 5 of MIL-HDBK-781D as a starting point). A vibration screen proof shall be performed to determine the optimum vibration level. The ANR units shall be tested for compliance with the operational requirements of this specification both before and after the vibration screen. If the final operational test is failed, the unit shall be repaired and rescreened. When the unit passes final operation, it shall then be thermally screened.

Thermal Screening: Prior to thermal screening a thermal survey shall be performed to determine the thermal response of the ANR modules to the thermal cycling of the chamber. Thermocouples shall be placed on the most thermally massive components of the ANR module and in chamber air. Controlled over-shooting of the chamber air beyond thermal limits of the hardware are permitted for the purpose of expediting the survey and subsequent screening. All ANR modules shall be submitted to thermal cycling of 20 cycles at a change rate of 20°C per minute measured at the component of the greatest thermal mass.

The storage temperature limits of MIL-E-5400 Class 1A are -57°C to +85°C. In keeping with these temperature limitations the temperature extremes for thermal cycling shall be stabilized at -55°C to +85°C. Thermal stability is considered to have occurred when the electronic part with the largest thermal mass or non-metallic portion of the PWB surface is within 5°C of the temperature range limits. The operational test shall be performed after the ANR has completed thermal screening. Failed ANR's shall be repaired and resubmitted to screening for 10 additional thermal cycles. Failed ANR's shall repeat this sequence no more than two times. In the event that an ANR fails the second failure free cycle, recommended corrective actions/alternative approaches shall be provided to the procurring activity.

5. PACKAGING

Packaging requirements for the ANR systems shall be as specified in the helmet specification, or purchasing document.

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6. NOTES:

This section contains information of a general explanatory nature that may be helpful, and explains rationale for requirements detailed in this specification. Information contained herein is not contractually binding.

6.1 General Description. The ANR system shall be a lightweight, airborne, solid state system integrated into the current rotary wing helmet earcups. The ANR system shall consist of the following assemblies:

a. Two earcup assemblies each containing the electronics of the ANR connected to a noise sampling microphone(s), and an earphone.
b. A helmet mounted environmentally sealed miniature toggle switch to manually bypass the ANR system allowing normal use of the earphones.
c. An electret boom microphone.
d. A helmet mounted audio frequency cable assembly to electrically connect the ANR system to the aircraft ICS cable assembly.

The ANR unit accepts audio signals and electrical power from the aircraft Intercommunication System (ICS), via the aircraft ICS audio frequency cable assembly. The ANR unit monitors the ambient acoustic noise present inside the helmet earcups, and electronically processes the acoustic noise signal with the ICS signal. The ANR unit emits an acoustic signal via the helmet earphones, which consists of the ICS signal and a noise signal out of phase with the ambient noise present in the earcups. This results in a minimization of the ambient acoustic noise level perceived at the ear of the aircrewman, without degradation of the ICS signal.

6.2 Intended use. Use of the ANR system will decrease the low frequency ambient noise sound pressure level to which an aircrewman is exposed, thereby decreasing the potential for hearing damage risk, and increasing the length of the mission time. During an aircraft power interruption, the ANR system will automatically switch to the bypass mode. In the bypass mode ICS audio signals are transferred directly to the earphones, without processing of the signal. Manual bypass of the ANR system is provided via a helmet mounted switch.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Number of ANR units required.
c. Level of packaging and packing required.
d. Special preparation for delivery requirements, if applicable (Section 5).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

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6.4.1 Conformity to qualified sample. It is understood that ANR systems supplied under contract shall be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection.

6.5 Application for qualification. The activity responsible for the QPL of this specification is the Naval Air Systems Command. The Naval Avionics Center, Indianapolis, Indiana, has been designated by the Naval Air Systems Command as agent for the establishment and maintenance of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

Commanding Officer
Naval Avionics Center
Code 444
6000 East 21st Street
Indianapolis, IN 46219-2189

6.6 Explaination of requirements. The following paragraphs provide the rationale for various requirements contained in this specification. This information is documented in order to prevent loss of the criteria and rationale used in developing the requirements stated herein. The information contained in the following paragraphs will provide a basis upon which a decision can be made regarding future requests for deviations to the requirements listed in this specification.

6.6.1 Earphone sensitivity and frequency response. A frequency response curve of the currently used H-87B/U earphones installed in an earcup of an SPH-3C helmet is shown in figure 1. The earcup was placed against the ear of a KEMAR manikin with a force of 1 Kg. It is desired that the ANR earphones provide the same level of performance or greater. Intelligibility is the goal of a communication system. Therefore acceptance criteria will be that specified in the intelligibility test rather than specified in a frequency response curve. However, the frequency response test will be performed to serve as a baseline from which performance can be measured when subjected to other tests specified herein.

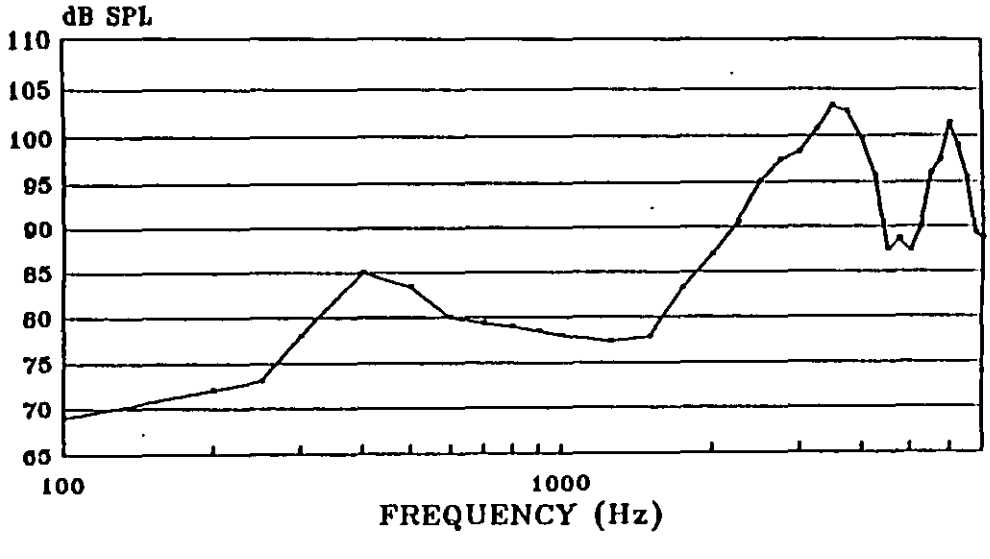


FIGURE 1. SPH-3C Earcup frequency response (kemar manikin).

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The artificial head method was chosen as the preferred test method due to the combination of repeatability and reliability test results. It was the opinion of the developing agency of this specification that greater accuracy in positioning of the earcup, relative to the measuring microphone, can be obtained since the earcup is removed from the helmet and can be visually inspected in the test configuration. The head size is constant, and therefore eliminates another variable when comparing ANR systems from vendors. This method will also permit the ANR system to be in an operational configuration when subjected to environmental conditions such as electromagnetic susceptibility testing, and temperature-altitude testing. Therefore data taken during environmental testing can be directly compared to data taken prior to these tests. This would be impossible if human subjects were used.

The rationale for establishing a specified dB SPL at frequencies 250 Hz and 400 Hz was to ensure that reproduction of the audible warning signals spec in 4.6.4.14 would not be degraded. The higher frequencies specified in 4.6.4.14 are in the speech range, and therefore would not be degraded since it would affect intelligibility.

6.6.1.1 Channel Balance. A difference of 4 dB was chosen to ensure deviation of earphone sensitivities would not be greater than that allowed for the currently used H-87B/U earphones, which has a sensitivity of 103 $\pm$ 2dB.

6.6.1.2 Passive attenuation. The minimum values required are representative of the attenuation provided by the SPH-3C helmet. These values are required in the event ANR would cease to operate. The requirements minimize the acceptance of poor helmet attenuation which may be possible if only total attenuation values were required.

6.6.2 Impedance. The input impedance of the ANR units is specified to be identical to the currently used H-87B/U earphones when in the bypass or active mode. The main goal is that the SPL emitted by the earphones of the ANR system is equal to that produced by the H-87B/U, and does not vary considerably as a function of frequency. A failure to meet the impedance requirement may be waived if the SPL at all frequencies exceeds that obtained from the H-87B/U, and the impedance is compatible with the control box C-2645 of the AN/AIC-14.

6.6.3 Stability. It is desired that the ANR unit remain stable during operation at all DC bias voltages. Therefore the voltage range specified in 3.4.2.11 was specified to be between the limits of 0 to 29 volts. A short or open circuit may occur at any of the input or output terminals of the ANR unit. Therefore the ANR unit should not be damaged or cause improper operation during or after these occurrences.

6.6.4 Distortion. Intelligibility is the main goal, and is performed using live subjects. However, intelligibility is not performed using the range of input signal levels for which the ANR system is expected to operate, since that will result in hearing discomfort for the test subject. For this reason, the distortion test is included. In order to provide an acceptable articulation loss for consonants in speech, the signal-to-noise (S/N) ratio should be 25 dB. The total Harmonic Distortion (THD) should be at least 10 dB below the S/N ratio to prevent addition of the THD to the noise. Therefore the THD should be -35 dB compared to the signal level. The percentage is calculated as follows:

$$\text{Percentage (P)} = 100 * 10^{(-35\text{dB}/20)}$$

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The percent contribution of the one-third octave bands between the limits of 300 Hz to 5000 Hz, to the articulation index is 90 percent. Above 5000 Hz, the contribution is minimal. Therefore the 300 Hz to 5000 Hz band was chosen. However, in order to be compatible with radar warning receivers, the frequency range was extended to 7 KHz. A distortion level of 1.8% was chosen for the bypass mode. For the active mode, a maximum distortion level of 2.5% was chosen to ensure the distortion would not exceed that currently allowed for the H-87B/U earphone.

6.6.5 Gain. It is desired that the performance of the ICS system will not be decreased when the ANR unit is switched from the active to bypass mode. When in the bypass mode, the ICS audio signal should be transferred directly to the earphones, with unity gain. Therefore the gain in the active mode should be unity to prevent a perceived degradation in ICS performance when switched from the active to bypass mode. A tolerance of +2 dB SPL is permitted.

6.6.6 Linearity. The upper limit of 0.82 watts was obtained by considering the maximum audio output voltage (3.5 volts RMS) of the interphone control box C-2645D into a 15 ohm load.

6.7 Subject term (key word) listing.

ACOUSTIC NOISE

ANR

AUDIO SIGNAL

INTERCOMMUNICATION NOISE

Preparing activity
Navy - AS
(Project 5965-N208)

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TABLE IV. Inspections.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.5.1
Group A (All sample units) 1/					
Soldering 1/	3.3.2.8	4.6.2.1	X	X	
Visual and Mechanical Inspection 1/	3.3.2.9	4.6.2	X	X	
Earphone Sensitivity 1/	3.4.1.1	4.6.3.1	X		X
Earphone Frequency Response 1/	3.4.1.2	4.6.3.2	X		X
Channel Balance	3.4.1.3	4.6.3.3	X		X
Linearity 1/	3.4.1.4	4.6.3.4	X		X
Overload Protection 1/	3.4.1.5	4.6.3.5	X		X
Active and Passive Attenuation 1/	3.4.1.6	4.6.3.6	X	X	
Intelligibility	3.4.1.7	4.6.3.7	X		

1/ Minimum requirements for QPL retention.

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TABLE IV. Inspections - Continued.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing
Undesired Sounds	3.4.1.8	4.6.3.8	X		X
Noise Floor	3.4.1.9	4.6.3.9	X		X
Audio Input Signal Voltage Range	3.4.2.1	4.6.4.1	X		X
Low Voltage Conditions 1/	3.4.2.3	4.6.4.3	X		X
DC Abnormal Operation 1/	3.4.2.4	4.6.4.4	X		X
Reverse Polarity Protection 1/	3.4.2.5	4.6.4.5	X		X
Power Consumption 1/	3.4.2.6	4.6.4.6	X	X	
Audio Modulation	3.4.2.7	4.6.4.7	X		X
Electrical Short Circuit Protection	3.4.2.8	4.6.4.8	X		X
Warm-up Time	3.4.2.9	4.6.4.9	X		X

1/ Minimum requirements for QPL retention

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TABLE IV. Inspections - Continued.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing
Impedance 1/	3.4.2.10	4.6.4.10	X	X	
Stability 1/	3.4.2.11	4.6.4.11	X		X
Distortion 1/	3.4.2.12	4.6.4.1	X		X
Gain 1/	3.4.2.16	4.6.4.13	X	X	
Group B					
Radiated and Conducted Susceptibility	3.4.2.15.1	4.6.4.12	X		
Group C					
Vibration 1/	3.4.3.1	4.6.5.1	X		
Temperature-Altitude 1/	3.4.3.2	4.6.5.2	X		
Humidity 1/	3.4.3.3	4.6.5.4	X		
Functional Shock	3.4.3.4.1	4.6.5.3.1	X		
Crash Hazard	3.4.3.4.2	4.6.5.3.2	X		

1/ Minimum requirements for QPL retention.

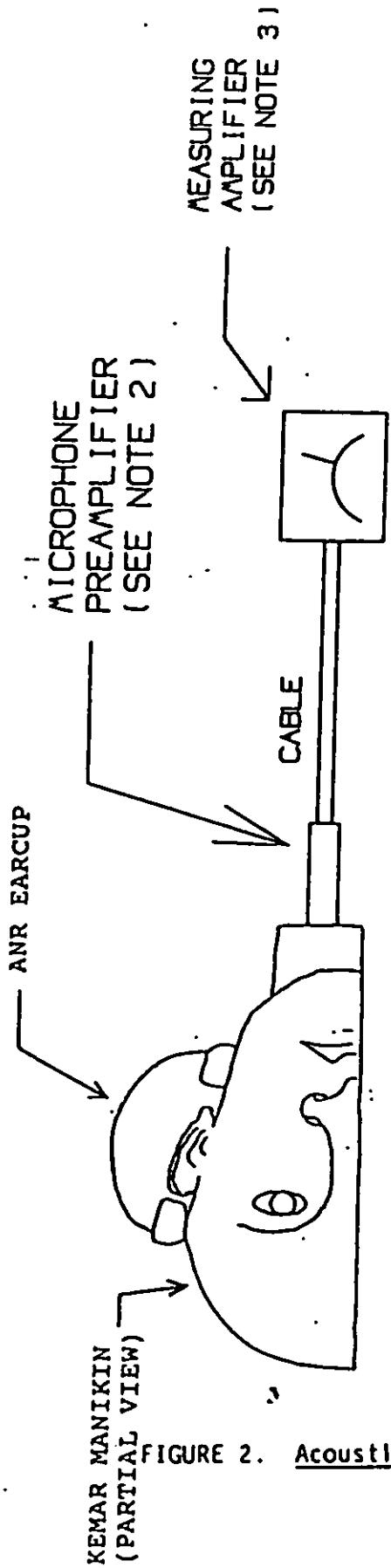
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TABLE IV. Inspections - Continued.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing
Acoustical Noise	3.4.3.5	4.6.5.5	X		
Fungus 1/	3.4.3.6	4.6.5.6	X		
Sand & Dust	3.4.3.7	4.6.5.7	X		
Salt Fog 1/	3.4.3.8	4.6.5.8	X		

1/ Minimum requirements for QPL retention.

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NOTE: THE FOLLOWING ARE SUGGESTED EQUIPMENT.

1. A KNOWLES ELECTRONICS MANIKIN FOR ACOUSTIC RESEARCH (KEMAR).
2. BRUEL & KJAER TYPE 2639 OR EQUIVALENT.
3. BRUEL & KJAER TYPE 2636, 2609, OR EQUIVALENT.
4. NOTE: SIGNAL GENERATORS & POWER SUPPLIES NOT SHOWN.
5. A ONE KILOGRAM WEIGHT SHALL BE PLACED UPON THE EARCUP.

FIGURE 2. Acoustical measurement test circuit arrangement.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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3. DOCUMENT TITLE

EARCUP, UNIT ACTIVE NOISE REDUCTION SYSTEM GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (*Include Zip Code*) SYSTEMS ENGINEERING & STANDARDIZATION DEPARTMENT, LAKEHURST,NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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